

ST115-HW2

James Holdaway

Packages

```
library(aRtsy)
```

Question #1

Install the aRtsy package. Show the code used to install the aRtsy package, but turn off the code chunk so it doesn't evaluate.

Next, explain the difference between `install.packages` and `library`

```
install.packages("aRtsy")
```

The code `install.packages("package-name")` installs the contents of a package to your computer. `library(package-name)` loads that package along with its functions, and data to your R session. In order to actually use a R package you must ensure that its been installed to your machine, then load it into the session using `library`.

Question #2

Create a level-1 section header at the top of your document and label it **Packages**. Next, under a **Question-2** level two section header, make a code chunk and load the **aRtsy** package. We are going to use a function in this package to plot the Collatz conjecture. On your own, research and explain what the Collatz conjecture is. Then, write the Collatz conjecture equation using in-line LaTeX code.

```
library(aRtsy)
```

The Collatz conjecture is a question that asks if you can bring any number down to one following one of two processes.

1. If even: $(n/2)$
2. If odd: $(3n + 1)$

This process repeats with n being redefined as the output until $n = 1$.

The conjecture can be represented like this:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$